```
(venv) (base) albina@Albinas-MacBook-Air scripts % python3 clean_data.py
Cleaning events.csv...
Done! Rows: 1274243
Cleaning campaigns.csv...
Done! Rows: 1907
Cleaning messages.csv...
Done! Rows: 2999153
Cleaning friends.csv and client_first_purchase_date.csv...
Done!
All data successfully cleaned and saved with '_cleaned.csv' suffix!
(venv) (base) albina@Albinas-MacBook-Air scripts % ls
```

```
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS
                                    NAMES
31086c4a0469   postgres:15    "docker-entrypoint.s…"   29 seconds ago   Up 28 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
                                    bd_postgres
0989c04588da   neo4j:latest   "tini -g -- /startup…"   29 seconds ago   Up 28 seconds   0.0.0.0:7474->7474/tcp, [::]:7474->7474/tcp, 7473/tcp, 0.0.0.0:76
87->7687/tcp, [::]:7687->7687/tcp   bd_neo4j
570be56394bd   mongo:latest   "docker-entrypoint.s…"   29 seconds ago   Up 28 seconds   0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
                                    bd_mongo

(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % head -n 1 data/messages.csv
id,message_id,campaign_id,message_type,client_id,channel,category,platform,email_provider,stream,date,sent_at,is_opened,opened_first_time_at,opened_last_
time_at,is_clicked,clicked_first_time_at,clicked_last_time_at,is_unsubscribed,unsubscribed_at,is_hard_bounced,hard_bounced_at,is_soft_bounced,soft_bounce
d_at,is_complained,complained_at,is_blocked,blocked_at,is_purchased,purchased_at,created_at,updated_at,user_device_id,user_id
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % head -n 1 data/client_first_purchase_date.csv
client_id,first_purchase_date,user_id,user_device_id
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % ./scripts/load_data_psql.sh
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
COPY 1907
COPY 2999153
COPY 1274243
COPY 1974238
COPY 174952
PostgreSQL data import finished successfully!

(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % ./scripts/load_data_mongodb.sh
Loading campaigns_cleaned.csv into collection 'campaigns'...
Successfully loaded 1907 documents into campaigns.
Loading messages_cleaned.csv into collection 'messages'...
/Users/albina/Desktop/big_data_assignment/scripts/load_data_mongodb.py:33: DtypeWarning: Columns (0: clicked_first_time_at, 1: clicked_last_time_at, 2: h
ard_bounced_at, 3: soft_bounced_at, 4: complained_at, 5: blocked_at, 6: purchased_at) have mixed types. Specify dtype option on import or set low_memory=
False.
  df = pd.read_csv(file_path)
Successfully loaded 2999153 documents into messages.
Loading events_cleaned.csv into collection 'events'...
Successfully loaded 1274243 documents into events.
Loading friends_cleaned.csv into collection 'friends'...
Successfully loaded 1974238 documents into friends.
Loading client_first_purchase_date_cleaned.csv into collection 'client_first_purchase'...
Successfully loaded 174952 documents into client_first_purchase.
MongoDB data import finished successfully!
MongoDB data import finished successfully!
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % pip install neo4j
Requirement already satisfied: neo4j in /Users/albina/venv/lib/python3.13/site-packages (6.1.0)
Requirement already satisfied: pytz in /Users/albina/venv/lib/python3.13/site-packages (from neo4j) (2026.1.post1)
```

```
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % python3 scripts/load_graph_fast.py
Importing Friends...
Importing Events...
Graph import finished!
```

```
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % python3 scripts/analyze_data.py
=== DATA ANALYSIS & BENCHMARKING ===

--- Q1: Campaign Analysis ---
Postgres Output: [('bulk', 2120975), ('transactional', 238330)]... | Avg: 268.71 ms | StdDev: 30.32 ms
MongoDB Output: [{'_id': 'bulk', 'total': 2120975}, {'_id': 'transactional', 'total': 238330}]... | Avg: 1355.41 ms | StdDev: 383.70 ms
Neo4j Output: [{'p.id': '3300443', 'freq': 1}, {'p.id': '3300323', 'freq': 1}]... | Avg: 21.07 ms | StdDev: 35.64 ms

--- Q2: Top Recommendations ---
Postgres Output: [('1004856', 13330), ('1004767', 12855)]... | Avg: 140.24 ms | StdDev: 24.69 ms
MongoDB Output: [{'_id': 1004856, 'views': 13330}, {'_id': 1004767, 'views': 12855}]... | Avg: 434.54 ms | StdDev: 61.37 ms
Neo4j Output: [{'p.id': '1004856', 'views': 41}, {'p.id': '1004767', 'views': 39}]... | Avg: 6.13 ms | StdDev: 4.66 ms

--- Q3: Full Text Search ---
Postgres Output: [('1005073', 'electronics.smartphone', 'samsung'), ('1004856', 'electronics.smartphone', 'samsung')]... | Avg: 0.26 ms | StdDev: 0.19 ms
MongoDB Output: [{'_id': ObjectId('69b84595513cfb61a9de2a91'), 'product_id': 1005073, 'category_code': 'electronics.smartphone'}, {'_id': ObjectId('69b84
595513cfb61a9de2a92'), 'product_id': 1004856, 'category_code': 'electronics.smartphone'}]... | Avg: 0.45 ms | StdDev: 0.29 ms
(venv) (base) albina@Albinas-MacBook-Air big_data_assignment % pip install matplotlib numpy
```